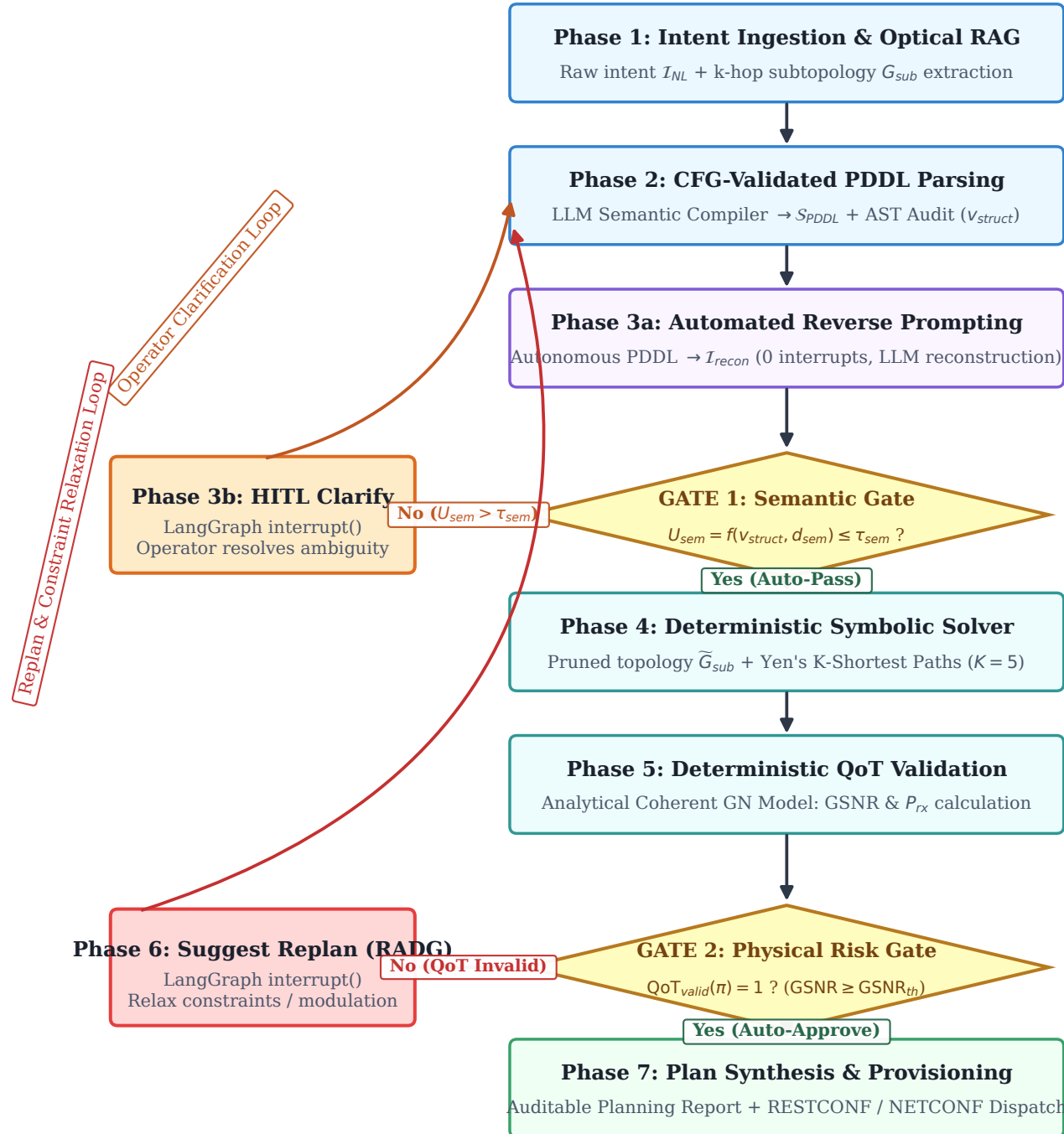




Fail-Fast Properties

Orthogonal Gating:

Decouples linguistic ambiguity (Gate 1) from physical propagation feasibility (Gate 2).

Zero-Overhead Autonomy:

When $U_{sem} \leq 0.30$ and QoT passes, execution completes with 0 human pauses.

Deterministic Physics:

GN model physics calculations never run on hallucinatory paths.

Protected Control Plane:

No unverified commands ever reach the SDN controller or ROADM hardware.

LangGraph Checkpointing:

Human-in-the-Loop interrupts use atomic state serialization.